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### As-needed filter cleaning

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#### Abstract

A method for controlling filter cleaning in a vacuuming device, the vacuuming device including at least one filter that is cleaned regularly and the filter cleaning including cleaning operations with individual cleaning pulses. An intensity of the cleaning operations can be set in dependence on a state of the filter of the vacuuming device for example by a variable length of time of a cleaning pulse and/or a variable number of cleaning pulses per cleaning operation. The method makes filter cleaning possible, in which the state of contamination or the aging of the filter in the vacuuming device can be taken into account. In a second aspect, the invention relates to a vacuuming device with which the proposed method can be carried out. For this purpose, the vacuuming device may for example comprise a suitable sensor system for detecting the state of the filter, and a control device.

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## Background/Summary

[0001] The present invention relates to a method for controlling filter cleaning in a vacuuming device, the vacuuming device comprising at least one filter that is cleaned regularly and the filter cleaning comprising cleaning operations with individual cleaning pulses.

### BACKGROUND

[0002] It is known that working with power tools on construction sites can generate dust. This dust can be picked up directly at its place of origin or at a place where it is deposited. For this purpose, vacuuming devices, such as dust extractors or vacuum cleaners, are usually used. In the context of the present invention, the term “construction-grade vacuuming device” is also preferably used. Many vacuuming devices or construction-grade vacuuming devices have filters or filter elements that can become clogged during the operation of the device. In order to clean the filters or filter elements, filter cleaning units are provided on some construction-grade vacuuming devices. In modern construction-grade vacuuming devices, systems that automatically clean the filter elements during operation are often used. When cleaning the filter, the vacuum in the vacuuming device that is used to suck in the dust is usually reduced, so that the sucking in of the dust is briefly interrupted and the filter is cleaned. In some devices, provision is made to flush the filter elements with a countercurrent and thus release any filter cake that may be present.

### SUMMARY OF THE INVENTION

[0003] The filter cleaning takes place in various construction-grade vacuuming devices by means of a brief backflush through the filter. This means that a brief reversal of the flow in the flow lines of the vacuum cleaner has the effect that for a short time an air flow can flow against the usual operating flow direction. This countercurrent can loosen or remove dust particles or a filter cake that may have formed on the filter. Backflushing is based on a pressure gradient that builds up across the filter during normal operation. The sudden actuation of a flap and/or opening in a flow line of the vacuum cleaner in the area of the suction turbine causes a brief reversal of the direction of flow through the filter during cleaning. The mentioned flap and/or opening in a flow line of the vacuum cleaner is preferably referred to in the context of the present invention as a “valve”.

[0004] The disadvantage of cleaning the filter by air backflushing is a pressure surge in the suction system caused by the air backflushing, which inevitably occurs and impairs the suction and thus the main function of the vacuuming device. Another disadvantage that is known in connection with such a pressure surge is that a cloud of dust can be blown out of the vacuuming device. In some conventional devices, such a dust cloud is created even if the suction flow is sufficiently throttled by a strong reduction in the negative pressure that is present in the vacuum cleaner in order to suck in the dust. Such an undesired emission of dust in the form of a dust cloud is to be avoided by the invention to be provided.

[0005] The filter cleaning process is usually made to take such a form that regularly recurring time intervals at which the filter cleaning is automatically carried out are stored or defined in the vacuuming device or in a control unit of the vacuuming device. In such time-based filter cleaning operations, cleaning cycles are carried out at regular time intervals of a constant length. In particular, in the case of conventional vacuuming devices known from the prior art, the opening times of the valves are of a constant length or are fixed by the manufacturer. Furthermore, in the case of many devices, the intervals between two individual pulses during a cleaning operation are of the same or essentially the same length.

[0006] However, the disadvantage of the filter cleaning methods known from the prior art with fixed time schedules is that the filter wear or the degree of contamination of the filter is not taken

into account when initiating the filter cleaning process, so that an essential factor for achieving a high cleaning quality is not included in the control of the cleaning. As a result, it can happen that cleaning takes place at unsuitable times. This means that the filter cleaning is for example carried out too early, and therefore too frequently, or too infrequently, and thus after a quality maximum with respect to the suction power has been exceeded. Too frequent cleaning is disadvantageous because the vacuum in the vacuuming device must be reduced during the cleaning, which causes a short-term, undesired interruption in the sucking in of the dust. This can have a detrimental effect on the suction power of the vacuuming device. Cleaning of the filter which is carried out too late or too infrequently is disadvantageous because this can give rise to periods in which the filter is already severely clogged, and therefore the suction power of the vacuuming device may be impaired. As a result, the effect of the filter may be reduced and more dust can get into the region of the turbine or blower. This in turn can lead to damage or impairment of the turbine and/or blower. In addition, the suction quality and suction efficiency of the vacuum cleaner can decrease.

[0007] An object of the present invention is to solve the problems described above and to provide a method for controlling filter cleaning in a vacuuming device with which as-needed filter cleaning is made possible and in which the design of a cleaning process takes into account the contamination of the filter. In particular, the invention is intended to reduce or even completely compensate for the pressure surge problem described at the beginning. A further object of the invention is to provide a vacuuming device which is set up to implement as-needed filter cleaning, in which a degree of contamination of the vacuum cleaner filter can be taken into account when performing the filter cleaning.

[0008] The present invention provides a method for controlling filter cleaning in a vacuuming device, the vacuuming device comprising at least one filter which is cleaned regularly, the filter cleaning comprising individual cleaning operations. The method is characterized in that an intensity of the cleaning operations is set in dependence on a state of the filter of the vacuuming device. In a preferred form taken by the invention, the intensity of the cleaning operations can be set in that cleaning pulses of the cleaning operations comprise lengths of time  $t_i$  which are set in dependence on a state of the filter of the vacuuming device. For the purposes of the invention, it is preferred that each cleaning operation comprises  $n$  cleaning pulses, each cleaning pulse being of a length of time  $t_i$ . The length of time  $t_i$  of the cleaning pulses per cleaning operation may preferably be set in dependence on a state of the filter of the vacuuming device. For the purposes of the invention, it may alternatively or additionally be preferred that a number  $n$  of cleaning pulses per cleaning operation can be set in dependence on a state of the filter of the vacuuming device. For the purposes of the invention, it is particularly preferred that the length  $t_i$  of the cleaning pulses and their number  $n$  per cleaning operation can be set independently of one another. In other words, it is for example preferred that consecutive cleaning operations have 2, 4, 3, 5 and again 3 cleaning pulses. The cleaning pulses may for example last 20, 58, 35, 47, 60, 92 or 108 ms. Preferably, any preferred number  $n$  per cleaning operation may be combined with any preferred pulse length  $t_i$ .

[0009] For the purposes of the invention, it is preferred that the cleaning pulses are in each case of an individual length of time  $t_i$ . The length of time  $t_i$  preferably corresponds to the time that a cleaning pulse takes, in other words the period of time that a cleaning pulse lasts. It can therefore preferably also be referred to as the "pulse length". A cleaning pulse  $i$  preferably takes a time  $t_i$ , for example a first cleaning pulse perhaps lasting a length of time  $t_1$ , while a second cleaning pulse is of a length of time  $t_2$ , etc. In particular, the letter "i" can be understood as an index for a period of time  $t$ . For the purposes of the invention, it is preferred that a cleaning operation consists of several cleaning pulses, which are respectively triggered by actuation of the valve.

[0010] It is preferred in the sense that cleaning operations take place at regular intervals during operation of the vacuuming device. The intervals between the individual cleaning operations may for example be in a range of around 15 s. However, other, shorter as well as longer, intervals are also conceivable. For the purposes of the invention, it is preferred that the intervals between the

individual cleaning operations may also be set in dependence on a state of the filter of the vacuuming device. The intervals between the individual cleaning operations preferably represent pauses between the cleaning operations in which normal suction operation of the vacuuming device can take place.

[0011] The cleaning pulses lead to the undesired pressure surges, the effects of which are advantageously compensated or minimized by the present invention. This is achieved for example in that the length of time  $t_i$  and/or a number  $n$  of cleaning pulses per cleaning operation can be set in dependence on a state of the filter of the vacuuming device, so that the intensity of a cleaning operation can be adapted to the respective state of the vacuuming device or its filter. For example, the proposed procedure makes it possible for the length of time  $t_i$  to be kept particularly short. In addition, the number of cleaning pulses can be kept as small as possible in order to also keep down the number of intrinsically undesirable pressure surges that can be caused by the cleaning pulses. Tests have shown that, with the proposed method, unnecessary excess cleaning can be avoided particularly effectively. In particular, the filter cleaning can be made to be particularly as needed, and therefore efficient, so that the filter cleaning or its intensity can be optimally adapted to the requirements of the current suction operation of the vacuuming device. In particular, an adaptive characteristic adjustment of filter cleaning during a cleaning operation in a vacuuming device can be made possible by the invention. For the purposes of the invention, it is preferred that a pressure level above and below the filter or a pressure level upstream and downstream of the filter of the vacuuming device in the direction of flow is determined and used as a basis for setting the length of time  $t_i$  or the number  $n$  of cleaning pulses. For the purposes of the invention, it is very particularly preferred that the pressure values are continuously measured at the different points within the vacuuming device and evaluated.

[0012] In the context of the proposed method, it is preferred that the pulse lengths  $t_i$  of the cleaning pulses of the cleaning operations can be made variable. The indices  $i$  are preferably between 1 and the number of cleaning pulses  $n$ , i.e.  $1 \leq i \leq n$ . For the purposes of the invention, it is very particularly preferred that each length  $t_i$  of a cleaning pulse  $i$  can be individually adapted to the filter state during the same cleaning operation. In other words, the lengths of the cleaning pulses within a cleaning operation may be different, depending on the state of the filter. If a cleaning operation has for example three cleaning pulses, the lengths of time  $t_1$ ,  $t_2$  and  $t_3$  can be set and carried out differently for each cleaning pulse. When determining the length of the cleaning pulses, the state of the filter of the vacuuming device is used as a basis in particular.

[0013] The situation that the filter is most heavily contaminated or clogged at the beginning of a cleaning operation is often found in vacuuming devices. It is then preferred for the purposes of the invention that a short pulse length  $t_1$  is set for the first cleaning pulse, while the pulse lengths  $t_2$ ,  $t_3$  etc. of the subsequent cleaning pulses 2, 3 etc. preferably last longer than the first cleaning pulse. As a result, excessive vigor of the first cleaning pulse can be effectively avoided and the intensity of the pressure surge generated during filter cleaning can be better distributed over multiple cleaning pulses. Frequently, with the first cleaning pulse of a cleaning operation, a large cloud of dust is undesirably blown out of the vacuuming device. Since the intensity of the cleaning pulses can be adapted by the proposed method to the current state of the filter or its degree of contamination, the emission of an initially large cloud of dust is avoided. Surprisingly, the more even distribution of the cleaning intensity over the multiple cleaning pulses, the length and/or number of which can be set in dependence on the state of the filter, reduces the total amount of dust clouds emitted per cleaning operation. This effectively avoids "back-pollution" of the environment and in particular prevents the user of a tool device with which the vacuuming device can be operated together from being exposed to high levels of dust.

[0014] In the context of the proposed method, it may also be preferred that the state of the filter of the vacuuming device does not only depend on its degree of contamination. It may also happen that the type of contamination or the material that clogs the filter has an influence on what form the

filter cleaning takes. For example, in the case of materials such as lime or gypsum, which do often cause heavy contamination of the filter, a small number of cleaning pulses can be set because the filter cake that forms is very dry and usually a small number of pressure surges is sufficient to remove the filter cake from the filter. In the case of moist filter cakes, on the other hand, it may be preferred that a higher number of cleaning pulses is set.

[0015] For the purposes of the invention, it is preferred that the length of time  $t_i$  of the cleaning pulses is in a range from 1 to 500 ms, preferably in a range from 10 to 200 ms, particularly preferably in a range from 30 to 150 ms and most preferably at approx. 40 to 100 ms. For example, pulse lengths  $t_i$  of 45 ms, 60 ms or 90 ms for a cleaning pulse may be very particularly preferred. Tests have shown that the times mentioned lead to optimum cleaning results.

[0016] For the purposes of the invention, it is preferred that the pulse length  $t_i$  is set by varying the valve opening time. In other words, different valve opening times can be set in order to achieve different periods of time  $t_i$ . The optimum valve opening time, and thus the optimum period of time  $t_i$ , are preferably determined by the control device of the vacuuming device. The control device preferably uses operating data and measured values that are determined by a suitable sensor system within the vacuuming device.

[0017] For the purposes of the invention, it is preferred that the valve opening time corresponds to the period of time in which the valve is open. For the purposes of the invention, it may be preferred that the beginning of the valve opening operation is considered in each case as the starting point of this period of time. However, it may also be preferred that the instant at which the valve is completely or essentially completely open, or is open with a predefined degree of opening, is used as the starting time. Analogously, it may be preferred that the beginning of the valve closing operation or the time at which the valve is completely or essentially completely closed again, or is closed with a predefined degree of closing, is used as the end time of the period of time. The same starting time and end time is preferably chosen for each cleaning pulse in order to carry out the proposed method. This makes it possible for comparable times or periods of time to be used for controlling the filter cleaning or for initiating the individual cleaning operations. For the purposes of the invention, it is preferred that a cleaning operation comprises a number of  $n$  cleaning pulses, the cleaning pulses being defined by the opening and closing of the valve. In particular, this also defines the lengths of time  $t_i$  of the cleaning pulses.

[0018] For the purposes of the invention, it is preferred that the opening time of the valve ("valve opening time") corresponds to the length of time  $t_i$  of a cleaning pulse. The valve opening time may correspond to the length of time  $t_i$ , i.e. it may be preferred for the purposes of the invention that the valve opening time is essentially identical to the length  $t_i$ . However, it may also be preferred that there is a fixed, known relationship between the valve opening time and the length of time  $t_i$  and that the length of time  $t_i$  can be calculated or derived from the valve opening time, or vice versa.

[0019] For the purposes of the invention, it is preferred that the valve opening time is varied using a control device. It may also be preferred that the intensity of a cleaning operation can be set with the aid of the control device. In other words, this preferably means that for example different valve opening times or cleaning pulse numbers can be set using software solutions. In this case, data relating to a state of the filter of the vacuuming device are preferably evaluated by the control device and used as a basis for determining an optimum valve opening time. For the purposes of the invention, it is preferred to refer to this form taken by the invention as a software setting of the length of time  $t_i$  or software setting of the valve opening time. For the purposes of the invention, it is preferred that the control device is part of the vacuuming device. Furthermore, the vacuuming device may comprise a suitable sensor system with which the data which describe the state of the at least one filter of the vacuuming device can be determined. This may be for example pressure sensors which are arranged within the flow lines in the interior of the vacuuming device. For example, pressure sensors may be arranged in a flow line upstream and downstream of a filter in

the direction of flow, so that the pressure conditions within the flow line upstream and downstream of the filter can be compared with one another. In particular, such an arrangement makes it possible that a pressure difference within the vacuuming device can be used as a control variable for the filter cleaning. The difference in the pressure conditions upstream and downstream of the filter can preferably be used to optimize the intensity of the cleaning operation. For this purpose, the determined pressure values may be used for example to set the pulse lengths  $t_i$  and/or the number  $n$  of cleaning pulses. In other words, the pressure conditions inside the vacuum cleaner, which may be determined for example upstream and downstream of the filter in the direction of flow, describe the state of the filter of the vacuuming device. These pressure conditions or their difference can then be used to control the intensity of the individual cleaning operations and to adapt them to the state of the filter. Tests have shown that the pressure difference is a particularly suitable parameter for the degree of contamination of the at least one filter of the vacuuming device. This data comparison or a previous processing or evaluation of the data may be carried out for example by the control device of the vacuuming device.

[0020] In the context of the invention, it is preferred that the data that are collected and evaluated for determining the pulse lengths  $t_i$  or for determining a number  $n$  of cleaning pulses are determined by sensors. They may be for example pressure sensors which are arranged upstream and downstream of the filter of the vacuuming device in the direction of flow. In the context of the invention, it is very particularly preferred to use in the determination of the pulse lengths  $t_i$  and/or the number  $n$  of cleaning pulses such data that are already collected for other reasons for the operation of the vacuuming device. In other words, it is preferred that the data that are used when carrying out the method to determine the lengths of time  $t_i$  or a number  $n$  of cleaning pulses are recorded with such a sensor system that is already present on the vacuuming device. As a result, the proposed method can be implemented in a vacuuming device in a particularly simple manner and without great effort. In particular, a software solution with which the proposed method is carried out makes it possible that vacuuming devices already on the market can be retrofitted particularly easily with the proposed method. In addition, it is economically advantageous if existing data are better utilized in order to improve the operation of the vacuuming device still further or to optimize the form taken by the filter cleaning.

[0021] For the purposes of the invention, it is preferred that a first sensor is arranged upstream of the filter of the vacuuming device in the direction of flow and that a second sensor is arranged downstream of the filter of the vacuuming device in the direction of flow. Preferably, a first pressure or a first pressure value prevailing upstream of the filter in the direction of flow is measured with the first sensor and a second pressure or a second pressure value prevailing downstream of the filter in the direction of flow is measured with the second sensor. The first pressure and the second pressure or the first pressure value and the second pressure value can be compared with one another and a pressure difference can be formed. This can be done for example by subtracting one of the two pressure values from the other pressure value. A person skilled in the art knows how to form a pressure difference from two pressure values. For the purposes of the invention, it is very particularly preferred that an intensity of the cleaning operations can be derived on the basis of the pressure difference, which is then set by suitable measures, such as optimizing the length of time  $t_i$  of cleaning pulses and/or optimizing the number  $n$  of cleaning pulses per cleaning operation. For the purposes of the invention, it is preferred that a required intensity of the cleaning operations to be carried out is determined on the basis of the determined pressure difference. In other words, it is preferred that the pulse lengths  $t_i$  of the subsequent cleaning pulses can be set on the basis of the determined difference in the pressure conditions that prevail upstream and downstream of the filter of the vacuuming device in the direction of flow. The determined pressure difference can preferably also be used to determine the number  $n$  of cleaning pulses. If there is a great need for cleaning, for example, more cleaning pulses may be carried out than if the need for cleaning is low. A high pressure difference preferably means that the filter has a high

degree of contamination. A high pressure difference preferably also leads to a strong pressure surge by the opening of the valve, so that the fact that the valve opening time can be set means that the strength of the cleaning pressure surge or a cleaning pulse or the volume of air flowing through the filter during a cleaning pulse can be optimally adapted to the degree of contamination of the filter, which is preferably related to the pressure difference.

[0022] For the purposes of the invention, it is preferred that the difference in the pressure conditions prevailing upstream and downstream of the filter of the vacuuming device in the direction of flow is a measure of the contamination of the filter. By means of filter cleaning adapted to the degree of contamination of the filter, in particular the number  $n$  of cleaning pulses can also be kept down. In this way, it can for example be avoided that too many cleaning pulses are carried out per cleaning operation. The avoidance of excessive cleaning pulses per cleaning operation represents a significant advantage of the invention, because it can preferably also keep down the number of undesirably ejected dust clouds. In addition, it can be ensured with the invention that each individual cleaning pulse has too great an intensity. This can preferably take place in that the valve opening times, the length of which preferably corresponds to the pulse length  $t_i$ , are kept short. For the purposes of the invention, it is preferred that a long cleaning pulse with a large time period  $t_i$  preferably corresponds to a high cleaning intensity, because a large volume of air is passed through the filter. For the purposes of the invention, it is preferred that the volume of air that is passed through the vacuum cleaner in the opposite direction of flow for filter cleaning is a measure of the intensity of filter cleaning. A short cleaning pulse with a short period of time  $t_i$  preferably corresponds to a low cleaning intensity, because in this case only a small amount of air or a small volume of air is passed through the vacuuming device.

[0023] In the context of the invention, it may alternatively be preferred that the intensity of the cleaning operation is influenced by a cross section of a valve opening that can be set. This form taken by the invention is preferably referred to as a mechanical adaptation of the filter cleaning. Different filter cleaning operations can preferably be achieved by the cross section of an opening and/or flap of the valve being changed. The fact that the cross section of the valve opening can be set means for the purposes of the invention in particular that a cross-sectional area within the valve or the flow line can be varied or set variably. This can be achieved for example by changing a valve opening path and/or a valve opening angle. These forms taken by the invention are preferably also referred to as an “adaptive valve opening displacement” or “adaptive valve opening angle”.

[0024] It may also be preferred for the purposes of the invention that the intensity of the cleaning operation can be influenced by setting a valve acceleration and/or valve damping. In other words, it may be preferred to vary the filter cleaning by adaptive valve acceleration or adaptive valve damping. Different cleaning intensities can be set in particular by setting different valve accelerations and/or valve dampings. For the purposes of the invention, the term “valve acceleration” preferably describes the dynamics on which the opening or closing process of the valve is based. A valve that closes at a high speed per unit of time has a high valve acceleration and a valve that closes at a low speed per unit of time has a low valve acceleration. In this connection, the term “speed” preferably describes the speed at which a valve flap moves when the valve that makes backflushing of the filter possible has such a valve flap. For the purposes of the invention, it is preferred to refer to this form taken by the invention as a “dynamic” setting of the cleaning intensity.

[0025] In a preferred form taken by the invention, the number  $n$  of cleaning pulses per cleaning operation is in a range from 1 to 7, preferably in a range from 2 to 5 and most preferably 3. Tests have shown that the numbers mentioned lead to particularly good cleaning results.

[0026] For the purposes of the invention, it is preferred that the number  $n$  of cleaning pulses can be set in dependence on the progress of the filter cleaning. In other words, for the purposes of the invention, it may be preferred to vary or set the number  $n$  of cleaning pulses such that a degree of contamination of the filter, a degree of clogging of the filter or an operating time of the vacuuming

device—for example measured since a last filter change—is taken as a basis or taken into account when establishing the number  $n$  of cleaning pulses. Adapting or being able to adapt the number  $n$  of cleaning pulses to the state of the filter allows particularly as-needed filter cleaning to be provided, with which on the one hand effective cleaning of the filter of the vacuuming device is made possible, but on the other hand also the pressure increase caused by the filter cleaning can be kept to a minimum or its duration can be kept as short as possible. As a result, the suction operation of the vacuum cleaner is impaired as little as possible and the suction efficiency remains high—in particular in comparison with conventionally working vacuum cleaners.

[0027] In a second aspect, the invention relates to a vacuuming device in the case of which the filter cleaning can be controlled by the proposed method. In particular, in the case of the proposed vacuuming device, a cleaning operation is made to take a form in dependence on a state of the filter of the vacuuming device. In other words, the proposed vacuuming device is set up to carry out the individual cleaning operations during operation of the vacuum cleaner in dependence on the state of its filter.

[0028] In particular, the proposed vacuuming device is advantageously not based on a time-based control with fixed intervals between cleaning operations, while the cleaning operations are also of a fixed length of time. In other words, the proposed vacuuming device does not work on the basis of rigid, fixed time schedules, but instead the operation of the vacuuming device or the implementation of the filter cleaning is based on the actual need for cleaning, so that the operation of the vacuuming device and the implementation of the filter cleaning can advantageously be adapted to the actual need for cleaning. In particular, in the case of the proposed vacuuming device, the number of cleaning pulses per cleaning operation is no longer fixed. Rather, the proposed invention allows the intensity of each cleaning operation to be set individually. This is made possible in particular by the fact that the pulse length  $t_i$  of each cleaning pulse  $i$  can be determined and set individually in dependence on a state of the filter which is arranged in the vacuuming device. In addition or as an alternative to this, the number  $n$  of cleaning pulses can be varied in dependence on a state of the filter. On the one hand, this avoids too frequent initiation of a cleaning process, which can have a detrimental effect on the performance of the vacuuming device. This is because in this case there are continual interruptions in the suction operation, since the filter cleaning is accompanied by an increase in pressure or a decrease in the negative pressure within the vacuuming device. In the case of a vacuuming device in which the filter cleaning is controlled according to the proposed method, the filter cleaning operations can advantageously be initiated in dependence on a current need and in dependence on the state of the filter. In this way, the number of filter cleaning processes per unit of time can be optimized so that on the one hand the vacuum cleaning process does not have to be interrupted too often and on the other hand an efficient suction power of the vacuuming device and an efficient filtering of the air flow in the vacuuming device can be ensured.

[0029] For the purposes of the invention, it is preferred that the vacuuming device comprises means for detecting a state of the at least one filter of the vacuuming device. The vacuuming device may for example comprise sensors for determining pressure values, it being possible in particular for the sensors to be arranged upstream and downstream of the filter in a suction-extraction stream of the vacuuming device. For the purposes of the invention, it may also be provided that the measured values determined with the sensors are further processed using information technology. For example, characteristic values, parameters or comparison data may be calculated and/or derived from the determined measured values.

[0030] The vacuuming device preferably also comprises a control device for carrying out the proposed method. In other words, it may be preferred for the purposes of the invention that the vacuuming device comprises a control device for adapting the length of time  $t_i$  and/or the number  $n$  of cleaning pulses per cleaning operation in dependence on a state of the filter of the vacuuming device. By adapting the length of time  $t_i$  and/or the number  $n$  of cleaning pulses, the intensity of a



cleaning operation can advantageously be set, preferably with the aid of the control device of the vacuuming device, in particular can be adapted to the degree of contamination of the filter of the vacuum cleaner.

[0031] The intensity of a cleaning operation can in particular also be adapted to the state of the filter during a cleaning operation. This adaptation can preferably take place at a point in time immediately before or during a cleaning pulse. It is preferred for the purposes of the invention that the intensity of the cleaning depends on the amount of air that flows through the filter of the vacuuming device in the opposite direction per unit of time. For the purposes of the invention, the term “in the opposite direction” preferably means that, during filter cleaning, the air or the air-dust mixture flows through the vacuum cleaner against the usual operating direction of flow. The intensity of an individual pulse preferably depends on the valve opening time, which is preferably linked to the length of time  $t_i$ . It is preferred for the purposes of the invention that the valve opening time has a significant influence on the intensity of the filter cleaning due to the provision of the air volume that can flow through the filter in the opposite direction. The amount of air or the volume of air that is available for backflushing the filter can preferably also be set by the cross-sectional area of the valve and/or its opening speed.

[0032] For example, this adaptation can take place in terms of software by setting the valve opening time. For example, it may be preferred that a computer program product with which the filter cleaning can be controlled is stored in the vacuuming device or in the control device of the vacuuming device. In particular, the computer program product can be used to set the valve opening times and thus the length of time  $t_i$  of the individual cleaning pulses. Furthermore, the intensity of the filter cleaning can be adapted by means of a cross section of the valve opening that can be set, such as for example by an adaptive valve opening displacement or an adaptive valve opening angle. In addition, the adaptation can be carried out by adaptive valve acceleration or adaptive valve damping. The number of cleaning pulses of a cleaning operation can preferably be adapted within limits according to the progress of the cleaning of the filter.

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## Description

### BRIEF DESCRIPTION OF THE DRAWINGS

[0033] In FIG. 1, the course over time of filter cleaning as known from the prior art is represented. Specifically:

[0034] FIG. 1 shows by way of example a course of filter cleaning as known from the prior art, represented on the basis of a plot of pressure  $p$  over time  $t$ .

[0035] In FIG. 2 a solely schematic vacuuming device according to the present invention is shown.

### DETAILED DESCRIPTION

[0036] FIG. 1 shows by way of example a course of filter cleaning according to a conventional filter cleaning method as known from the prior art. In FIG. 1, the plot of pressure  $p$  over time  $t$  is represented. In the plot represented in FIG. 1, the pressure  $p$  inside the vacuuming device (not shown) is plotted on the y axis, while the time  $t$  is plotted on the x axis. Thus, FIG. 1 preferably shows a change over time in the pressure conditions within the vacuuming device during its operation, in which a filter cleaning operation (2) takes place.

[0037] In the excerpt from the operation of a vacuuming device that is represented in FIG. 1, a cleaning operation (2) is shown. The cleaning operation (2) shown comprises three cleaning pulses (3), which are respectively represented in the plot as a peak. The cleaning pulses (3) are in each case of a length of time  $t_i$  (1), the first cleaning pulse (3) being of a length of time  $t_1$ , while the second cleaning pulse (3) is of a length of time  $t_2$  and the third cleaning pulse (3) is of a length of time  $t_3$ . In the example shown in FIG. 1, the lengths of time  $t_i$  are the same or essentially the same. In this respect, the pressure profile over time as represented in FIG. 1 corresponds to the pressure

profile in a conventional vacuum cleaner with a constant length  $t$  of a cleaning pulse (3) and a fixed number of cleaning pulses (3).

[0038] FIG. 2 shows a vacuuming device 100 performing the filter cleaning of the present invention in which an intensity of the cleaning operations is set in dependence on a state of the filter 102 of the vacuuming device. The vacuuming device 100 includes a sensor 104 for detecting the state of the filter and a controller 108 for adapting a length of time or a number  $n$  of cleaning pulses per cleaning operation in dependence on the state of the filter by opening or closing a valve 106, and for setting cross section of a valve opening of valve 106 and/or setting a valve acceleration or valve damping.

#### LIST OF REFERENCE SIGNS

[0039] 1 length of time  $t_i$  [0040] 2 cleaning operation [0041] 3 cleaning pulse

## Claims

1-13. (canceled)

14: A method for controlling filter cleaning in a vacuuming device, the vacuuming device including at least one filter cleaned regularly by cleaning operations, the method comprising: setting an intensity of the cleaning operations in dependence on a state of the filter of the vacuuming device.

15: The method as recited in claim 14 wherein the intensity is set by setting lengths of time of cleaning pulses of the cleaning operations.

16: The method as recited in claim 15 wherein the lengths of time are set by varying a valve opening time.

17: The method as recited in claim 16 wherein the valve opening time is varied using a control device.

18: The method as recited in claim 15 wherein the lengths of time of the cleaning pulses are in a range from 1 to 500 ms.

19: The method as recited in claim 18 wherein the range is from 30 to 500 ms.

20: The method as recited in claim 18 wherein the range is from 40 to 100 ms.

21: The method as recited in claim 14 wherein the intensity is set by setting a number  $n$  of cleaning pulses per cleaning operation.

22: The method as recited in claim 21 wherein the number  $n$  is in a range from 1 to 7.

23: The method as recited in claim 21 wherein the number  $n$  is in a range from 2 to 5.

24: The method as recited in claim 21 wherein the number  $n$  is 3.

25: The method as recited in claim 21 wherein the number  $n$  of cleaning pulses per cleaning operation is set in dependence on progress of the filter cleaning.

26: The method as recited in claim 14 wherein the intensity of the cleaning operations is influenced by a settable cross section of a valve opening.

27: The method as recited in claim 14 wherein the intensity is influenced by setting a valve acceleration or valve damping.

27: A vacuuming device performing the filter cleaning as recited in claim 14.

28: The vacuuming device as recited in claim 27 comprising a sensor for detecting the state of the filter.

29: The vacuuming device as recited in claim 27 comprising a controller for adapting a length of time or a number  $n$  of cleaning pulses per cleaning operation in dependence on the state of the filter.

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